

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2020

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Time Allowed: 3 Hours

Teacher's Name: Mr. Abdullah Iqbal

Date: 15-JAN-2021

Note: Please attempt five questions. All questions carry equal marks.

Q.1 (a) How could you define Poisson distribution as the limiting form of Binomial distribution? **(02)**

(b) Before buying a batch of 100 fuses, a man tests 4 of these. If the batch contains 10 defective fuses, what is the probability that he will find **(i)** no defective fuse in the sample? **(ii)** 4 defectives in a sample? **(04)**

(c) Given that the height of college boys has an average of 5'-2" and standard deviation of 4", and that the minimum height required for joining the N.C.C. is 5'-4", find the percentage of boys who would be rejected on account of their height? **(04)**

Q.2 (a) How could you define Binomial distribution as an approximation of Normal Distribution? **(02)**

(b) The international coffee organization reported that percentage people over age of 10 who drink decaffeinated coffee increased from 4% in 1960 to 15% 1985. Suppose in 1985 a random sample of 5 persons over 10 is asked if they drink decaffeinated coffee. What is the probability that at least two will answer yes? **(04)**

(c) A man goes by car to his office, and route through city center takes him, on average, 27 minutes with a standard deviation of 5 minutes. With the opening of a new ring road, the man can bypass the congestion of the city center, but journey now takes, on the average, 29 minutes with a standard deviation of 2 minutes. Determine which route is the better one if the man has **(i)** 28 minutes, and **(ii)** 32 minutes to reach his office for an appointment. **(04)**

Q.3 (a) Explain clearly the concepts of Random Variable. What is a distribution function? **(03)**

(b) The following table shows the joint probability distribution of random variables X and Y **(04)**

Y \ X	X	1	2
	1	2/15	1/15
	3	4/15	1/15
	5	3/15	4/15

Compute the conditional distribution $f(x/y)$. Check whether X and Y are independent random variables.

(c) For what value of A, the function defined as below will be a p.d.f **(03)**

$$f(x) = \begin{cases} Ax^3(1-x), & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Also find its mean and variance.

Q.4 (a) Describe a Poisson Distribution with the help of at least two real life examples? **(02)**

(b) Suppose that airplane engines operate independently and fail with probability of 1/5. If a plane can make a successful flight with half of its engines working determine whether a 4-engine plane or a 2-engine plane has a chance of making a successful flight. **(04)**

(c) A manufacturer of cotter pins knows that 5 percent of his product is defective. If he sells cotter pins in boxes of 100, and guarantees that not more than 4 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality? **(04)**

Q.5 (a) How could you define Normal Distribution in relation with its parameters? **(02)**

(b) An irregular six-faced die is thrown and the expectation that in 10 throws it will give five even numbers is twice the expectation that it will give four even numbers. How many times, 10,000 sets of 10 throws would you expect it to give no even number? **(04)**

(c) A random sampling of 4 members club has shown that 3 prefer no smoking in the clubhouse dining room. What is the probability that this will occur if in fact only 20% of members prefer no smoking in the dining room. **(04)**

BEST OF LUCK